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HSL

HALF YEARLY EXAMINATION - 2025
PHYSICS

12 - Std
Time : 3.00 Hrs

Makrs : 70

15X1=15

I **CHOOSE THE BEST ANSWER :**

1. The energy stored in a coil of inductance 5H and resistance 20Ω when the emf applied to the coil is 100 volt
(a) 62.5 J (b) 125 J (c) 12.5 J (d) 15.6 J

2. The variation of frequency of carrier wave with respect to the amplitude of the modulating signal is called-
(a) Amplitude modulation (b) Frequency modulation
(c) Phase modulation (d) Pulse width modulation

3. A thin insulated wire forms a plane spiral of N = 100 tight turns carrying current I = 8 m A (milli ampere). The radii of inside and outside turns are a=50 mm and b =100 mm respectively. The magnetic induction at the center of the spiral is
(a) 5 μT (b) 7 μT (c) 8 μT (d) 10 μT

4. In hydrogen atom, which of the following transitions produce spectral line of maximum frequency?
a) 2 → 1 b) 6 → 2 c) 4 → 3 (d) 5 → 1

5. Consider an oscillator which has a charged particle and oscillates about position with a of 30 MHz. The wavelength of electromagnetic waves produced by this oscillator is
(a) 1 m (b) 10 m (c) 100 m (d) 1000 m

6. When light is incident on a soap film of thickness 5×10^{-5} cm, the wavelength of light reflected maximum in the visible region is 5320 A^0. Refractive index of the film will be,
(a) 1.22 (b) 1.83 (c) 1.51 (d) 1.33

7. The nucleus is approximately spherical in shape. Then the surface area of nucleus having mass number A varies as
(a) $A^{\frac{2}{3}}$ (b) $A^{\frac{4}{3}}$ (c) $A^{\frac{1}{3}}$ (d) $A^{\frac{5}{3}}$

8. When a telescope is in normal adjustment, the distance of the objective from the eyepiece is found to be 100 cm. If the magnifying power of the telescope at normal adjustment is 24, the focal length of the lenses are
(a) 96 cm, 4 cm b) 48 cm, 2 cm c) 50 cm, 50 cm d) 80 cm, 20 cm

9. An electric field $E=10x\hat{i}$ exists in a certain region of space. Then the potential difference $V = V_0 - V_A$, (where V_0 is the potential at the origin and V_A is the potential at x = 2 m) is
(a) 10 V (b) +20 V (c) - 20 V (d) -10 V

10. In an oscillating LC circuit, the maximum charge on the capacitor is Q The charge on the capacitor when the energy is stored equally between the electric and magnetic fields is
(a) $\frac{Q}{\sqrt{2}}$ (b) $\frac{Q}{\sqrt{3}}$ (c) $\frac{Q}{2}$ (d) Q

11. Two radiations with photon energies 0.9 eV and 3.3 eV respectively are falling on a metallic surface successively. If the work function of the metal is 0.6 eV, then the ratio of maximum speeds of emitted electrons will be
a) 1:4 b) 1:9 c) 1:1 d) 1:3

12. The gravitational waves were theoretically proposed by
(a) Conrad Rontgen (b) Marie Curie (c) Albert Einstein (d) Edward Purcell

13. The radius of curvature of curved surface at a thin plano convex lens is 10cm and the refractive index is 1.5. If the plane surface is silvered, then the focal length will be,
(a) 5 cm (b) 15 cm (c) 10 cm (d) 20 cm

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14. A carbon resistor of (47 ± 4.7)k Ω to be marked with rings of different colours for its identification. The colour code sequence will be
a) Yellow – Green - Violet - Gold
b) Violet- Yellow - Orange – Silver
c) Yellow - Violet - Orange – Silver
d) Green – Orange - Violet - Gold

15. If an electric field is given $10\hat{i}+3\hat{j}+4\hat{k}$ calculate the electric flux through a surface of area 10 units lying in yz plane
a) 100 units b) 10 units c) 30 units d) 40 units

II. ANSWER ANY SIX QUESTIONS : QUESTION NO. 24 COMPULSORY: 6 X 2 = 12

16. State Kirchhoff's voltage rule.
17. Why are e.m. waves non-mechanical?
18. Define Curie.
19. Dielectric strength of air is 3×10^6 V m^{-1}. Suppose the radius of a hollow sphere in the Van de Graff generator is R = 0.5 m, calculate the maximum potential difference created by this Van de Graaff generator.
20. A proton and an electron have same kinetic energy. Which one has greater de Broglie wavelength. Justify.
21. How will you define Q-factor?
22. What do you mean by doping?
23. What are grating elements? Write its unit.
24. An object is placed at a certain distance from a convex lens of focal length 20 cm. Find the distance of the object if the image obtained is magnified 4 times.

III. ANSWER ANY SIX QUESTIONS : QUESTION NO. 33 COMPULSORY: 6 X 3 = 18

25. Derive an expression for electrostatic potential due to a pointcharge.
26. Find out the phase relationship between voltage and current in a pure resistor circuit.
27. Find the magnetic field due to a long straight conductor using Ampere's circuital law.
28. List out the characteristics of photons.
29. A person has farsightedness with the minimum distance he could see clearly is 75 cm. Calculate the power of the lens of the spectacles necessary to rectify the defect.
30. Draw the circuit diagram of a half wave rectifier and explain its working.
31. Explain the determination of the internal resistance of a cell using potentio meter.
32. Obtain the expression for apparent depth.
33. Compute the binding energy of ^4_2He nucleus using the following data: Atomic mass of Helium atom, $M_A(He) = 4.00260u$ and that of hydrogen atom , $m_H = 1.00785u$.

IV. ANSWER ALL THE QUESTIONS: 5 X 5 = 25

34. (A)What is emission spectrum? Explains its types. ***(OR)***
(B) Describe the function of a transistor as an amplifier with the neat circuit diagram. Sketch the input and output wave forms.
35. (A) Explain the working of a single-phase AC generator with necessary diagram. ***(OR)***
(B) Obtain lens maker's formula and mention its significance.
36. (A) Calculate the electric field due to a dipole on its equatorial plane. ***(OR)***
(B) Discuss the Millikan's oil drop experiment to determine the charge of an electron.
37. (A)(i)Define temperature coefficient of resistivity.
(ii)The resistance of a nichrome wire at 20°C is 10 Ω.If its temperature coefficient of resistivity of nichrome is 0.004/°C, find the resistance of the wire at boiling point of water. Comment on the result. ***(OR)***
(A) (i) Define work function of a metal.Give its unit.
(ii) A radiation of wave length 300 nm is incident on a silver surface. Will photoelectrons be observed? (work function of silver= 4.7 eV)
38. (A) Derive the expression for the force between two parallel, current-carrying conductors.
(OR) (B) Obtain the equation for bandwidth in Young's double slit experiment.

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